

Confidence Intervals:

Def. (Confidence Interval): Let θ be an unknown parameter, and $X_1, \dots, X_n$ a random sample. For error level $\alpha \in (0, 1)$, we say that $[\underline{\theta}(X_1, \dots, X_n), \bar{\theta}(X_1, \dots, X_n)]$ is a $1-\alpha$ confidence interval for θ if:

$$P(\underline{\theta}(X_1, \dots, X_n) \leq \theta \leq \bar{\theta}(X_1, \dots, X_n)) = 1 - \alpha.$$

i.e., " θ lies outside $[\underline{\theta}, \bar{\theta}]$ with probability α "

Note: The interval $[\underline{\theta}(X_1, \dots, X_n), \bar{\theta}(X_1, \dots, X_n)]$ is random here, not the parameter θ . This differs from our usual statements about "P(a R.V. is between two constants)".

Note: Looks similar to percentiles / critical values, but here θ is not random. Still, they will help us!

Note: A tradeoff!

Smaller $\alpha \Leftrightarrow$ Wider Interval

Larger $\alpha \Leftrightarrow$ Narrower Interval

Common choices of α are 10%, 5%, 1%.

To build confidence intervals (LI's), we need to know something about distributions - specifically of $\underline{\theta}(X_1, \dots, X_n)$ and $\bar{\theta}(X_1, \dots, X_n)$.

$\leadsto$ The best tool we have here is the Central Limit Theorem.

Ex.: Let $X_1, \dots, X_n$ be iid from some unknown population with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$, where μ (but not σ^2) is unknown. Assume $n > 30$.

Find a $1-\alpha$ C.I. for μ .

By CLT, $\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \overset{\text{approx}}{\sim} N(0,1)$. || Note: the z-score of $\bar{X}_n$ is approx. $N(0,1)$ by CLT!

By def'n of critical values, then:

$$P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z_{\alpha/2}\right) \approx 1 - \alpha$$

Want to isolate μ here

$$P\left(-\sigma z_{\alpha/2} \leq \sqrt{n}(\bar{X}_n - \mu) \leq \sigma z_{\alpha/2}\right) \approx 1 - \alpha \quad \text{multiplying by } \sigma > 0$$

$$P\left(-\frac{\sigma}{\sqrt{n}} z_{\alpha/2} \leq \bar{X}_n - \mu \leq \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) \approx 1 - \alpha \quad \text{dividing by } \sqrt{n}$$

$$P\left(-\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \leq -\mu \leq -\bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) \approx 1 - \alpha \quad \text{subtracting } \bar{X}_n$$

$$P\left(\bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \geq \mu \geq \bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) \approx 1 - \alpha \quad \text{multiplying by } -1$$

$$\therefore P\left(\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \leq \mu \leq \bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) \approx 1 - \alpha$$

$\underbrace{\bar{X}_n}_{\text{"}\bar{\theta}(X_1, \dots, X_n)\text{"}} \quad \underbrace{\mu}_{\text{"}\theta\text{"}} \quad \underbrace{\bar{X}_n}_{\text{"}\bar{\theta}(X_1, \dots, X_n)\text{"}}$
 $\uparrow \text{random!} \quad \uparrow \text{parameter of interest} \quad \uparrow \text{random!}$

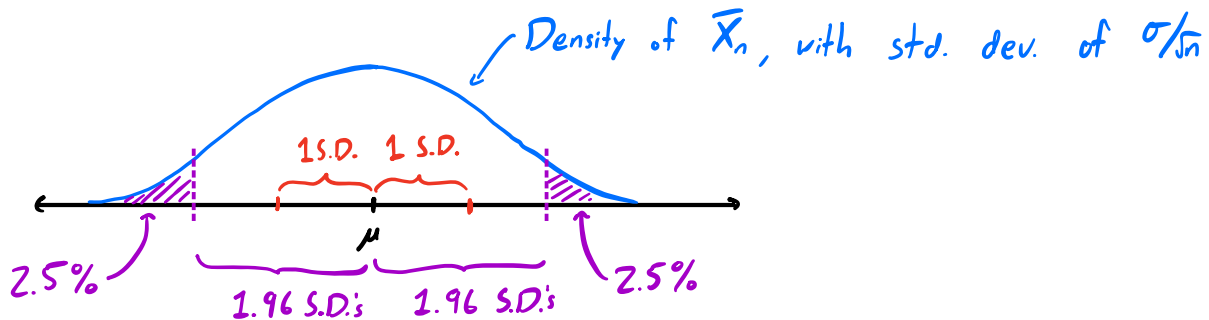
$\Rightarrow \left[\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \right]$ is an approximate $1-\alpha$ Confidence Interval for μ .

(Note: the above is sometimes written " $\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$ ")

Intuition:

$X_1, \dots, X_n$ iid, with unknown mean μ , known variance σ^2 .

Let $\alpha = 5\%$, and $z_{\alpha/2} = z_{2.5\%} = 1.96$



Hence: The probability that $\bar{X}_n$ is within 1.96 standard deviations ($1.96 \cdot \frac{\sigma}{\sqrt{n}}$) of its mean μ is 95%.



The probability that μ is within 1.96 std. dev.'s ($1.96 \cdot \frac{\sigma}{\sqrt{n}}$) of $\bar{X}_n$ is 95%.

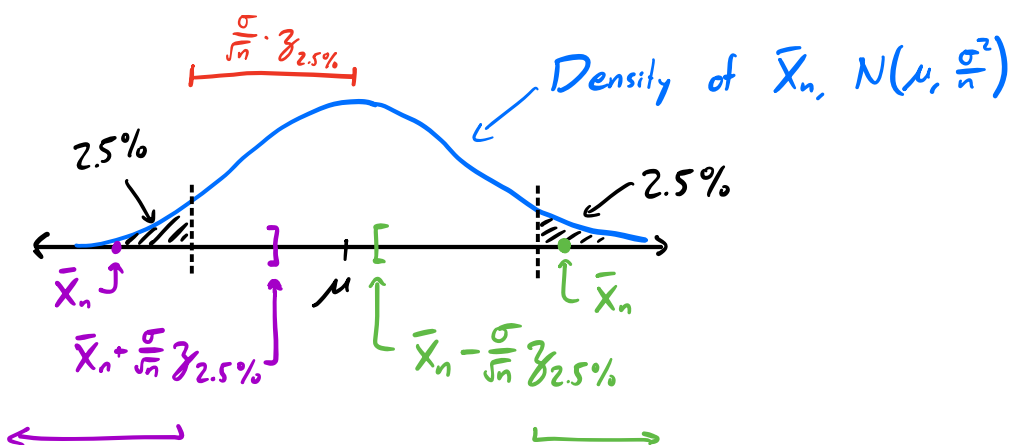
The same statement!

~> Now this is an inferential statement!

" μ within $1.96 \cdot \frac{\sigma}{\sqrt{n}}$ of $\bar{X}_n$ " $\Leftrightarrow$ " $\mu \in \bar{X}_n \pm 1.96 \frac{\sigma}{\sqrt{n}}$ "

exactly our 95% C.I. for μ !

Illustration:



Prob. of this is 2.5% || If $\bar{X}_n$ falls here, then $\bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{2.5\%} \leq \mu$

If $\bar{X}_n$ falls here, $\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{2.5\%} \geq \mu$ || Prob. of this is 2.5%

$\Rightarrow \mu$ is not in

$$\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{2.5\%}$$

$\Rightarrow \mu$ is not in

$$\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{2.5\%}$$

$$\therefore P(\mu \notin \bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{2.5\%}) = 2.5\% + 2.5\%$$

$$\Leftrightarrow P(\mu \in \bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{2.5\%}) = \underline{\underline{95\%}}$$

Obs. #1: The $1-\alpha$ CI of $\bar{X}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$ has a fixed width of $2 \cdot \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$, but is centered at the random value $\bar{X}_n$.

If:

$$\text{Large } \alpha \Rightarrow \text{Small } z_{\alpha/2} \Rightarrow \begin{cases} \text{High Precision (Low Width)} \\ \text{but} \\ \text{Low Reliability (Prob. of containing } \mu) \end{cases}$$

$$\text{Small } \alpha \Rightarrow \text{Large } z_{\alpha/2} \Rightarrow \begin{cases} \text{High Reliability} \\ \text{but} \\ \text{Low Precision} \end{cases}$$

The width also increases in σ , and decreases in n .

Obs. #2: This $1-\alpha$ CI was derived beginning from:

$$P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z_{\alpha/2}\right) \simeq 1 - \alpha$$

But it is also true that:

$$\bullet P\left(-z_{\alpha} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}\right) \simeq 1 - \alpha$$

$$\bullet P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z_{\alpha}\right) \simeq 1 - \alpha$$

$$\text{And even } P\left(-z_{\alpha/4} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \leq z_{3\alpha/4}\right) \simeq 1 - \alpha$$

Each of these could be used to build a different $1-\alpha$ CI for μ !

Hence, $1-\alpha$ CI's are not unique!

Prop. ($1-\alpha$ CI's for pop. mean - known variance, $n > 30$):

Let $X_1, \dots, X_n$ be iid from some distribution with unknown mean μ and known variance $\sigma^2 < \infty$. If $n > 30$:

- $[\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}]$ is a two-sided $1-\alpha$ CI for μ .
- $[\bar{X}_n - \frac{\sigma}{\sqrt{n}} z_{\alpha}, \infty)$ is a lower one-sided $1-\alpha$ CI for μ .
- $(-\infty, \bar{X}_n + \frac{\sigma}{\sqrt{n}} z_{\alpha}]$ is an upper one-sided $1-\alpha$ CI for μ .

Note: Once we observe realizations $x_1, \dots, x_n$, a two-sided $1-\alpha$ CI is now $\bar{x}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$. This interval is no longer random!

Either $\mu \in \bar{x}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$, or $\mu \notin \bar{x}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$.

So it is not correct to say $P(\mu \in \bar{x}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}) = 1-\alpha$.

Instead, we say that we are $(100 \cdot (1-\alpha))\%$ confident that

$\bar{x}_n \pm \frac{\sigma}{\sqrt{n}} z_{\alpha/2}$ contains μ .

Sampling & CI's:

Imagine we did the following:

- Fix n , and some distribution for iid $X_1, \dots, X_n$

- For $i = 1 \dots 1000$:

- Generate realizations $x_1, \dots, x_n$

- Check if $\mu \in \bar{x}_n \pm 1.96 \frac{\sigma}{\sqrt{n}}$

- Return: (# of i 's where μ is in the CI) / 1000

} This will approx. equal the confidence level, here 95%.

Two very important assumptions were made so far:

i) The pop. variance σ^2 is known

ii) $n > 30$, so the CLT approximation is acceptable

In order to build CI's without either (or both) of these assumptions, we will need some more CLT-like results.

Unknown Variance:

It is unrealistic to assume the pop. variance is known, but the mean is not.

For instance, if $X \sim \text{Poi}(\lambda)$, then $E[X] = \lambda = \text{Var}(X)$. If we knew the variance of X , we'd also know its mean!

Thm. (CLT w/ unknown variance): Let $X_1, \dots, X_n$ be iid, with both $\mu = E[X_i]$ and $\sigma^2 = \text{Var}(X_i) < \infty$ unknown. If $n > 40$, then:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{s(X_1, \dots, X_n)} \underset{\text{approx}}{\sim} N(0, 1),$$

$$\text{where } s^2(X_1, \dots, X_n) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Note: The main difference between this and the original CLT is that we replace the (now unknown) variance σ^2 with a point estimator of the variance - given by $s^2(X_1, \dots, X_n)$.

Since we have to replace the (now unknown) variance σ^2 with a random estimator $s^2(X_1, \dots, X_n)$, we will need to have a larger sample size for the CLT's normal approximation to be passable.

We can now use this variation of the CLT to build confidence intervals when the population variance is unknown!

Corollary (CI's for pop. mean - unknown variance, $n > 40$): Let $X_1, \dots, X_n$ be iid from some distribution with unknown mean μ and variance $\sigma^2 < \infty$.

If $n > 40$

• $[\bar{X}_n - \frac{\hat{S}}{\sqrt{n}} z_{\alpha/2}, \bar{X}_n + \frac{\hat{S}}{\sqrt{n}} z_{\alpha/2}]$ is a two-sided $1-\alpha$ CI for μ .

• $[\bar{X}_n - \frac{\hat{S}}{\sqrt{n}} z_{\alpha}, \infty)$ is a lower one-sided $1-\alpha$ CI for μ .

• $(-\infty, \bar{X}_n + \frac{\hat{S}}{\sqrt{n}} z_{\alpha}]$ is an upper one-sided $1-\alpha$ CI for μ .

where $\hat{S}^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$.

Proving this is almost identical to the case of CI's with known variance. Only that we use the latest version of the CLT.

Note: Here, the width of my two-sided $1-\alpha$ CI is equal to $2 \cdot z_{\alpha/2} \frac{\hat{S}}{\sqrt{n}}$, which is now random, since $\hat{S}^2$ depends on $X_1, \dots, X_n$.

Example: Polling

A population has voters who either support candidate A or candidate B for an upcoming election.

Suppose we poll 100 individuals from this population, and observe that 54% of them support candidate A.

Goal: Find a two-sided 95% CI for the proportion of voters that support A.

In Math: $X_1, \dots, X_{100} \stackrel{iid}{\sim} \text{Ber}(p)$. We observed $\bar{x} = \frac{1}{100} \sum_{i=1}^{100} x_i = 0.54$.

$$\begin{aligned} \text{Compute: } s^2(x_1, \dots, x_{100}) &= \frac{1}{100-1} \sum_{i=1}^{100} (x_i - \bar{x})^2 \\ &= \frac{1}{99} \sum_{i=1}^{100} x_i^2 - \frac{100}{99} \cdot \bar{x}^2 \quad \text{Some algebra} \\ &= \sum_{i=1}^{100} x_i, \text{ since } 1^2=1, 0^2=0, \text{ and } x_i \in \{0, 1\} \\ &= 100 \cdot \bar{x} \end{aligned}$$

$$\begin{aligned} s^2(x_1, \dots, x_{100}) &= \frac{100}{99} (\bar{x} - \bar{x}^2) \\ s^2(x_1, \dots, x_{100}) &\approx 0.25 \quad \bar{x} = 0.54 \end{aligned}$$

Hence: A 95% CI for p is $\left[\overbrace{0.54}^{\bar{x}} - \overbrace{1.96}^{z_{2.5\%}} \cdot \overbrace{\sqrt{\frac{0.25}{100}}}^{\sqrt{s^2/100}}, \overbrace{0.54}^{\bar{x}} + \overbrace{1.96}^{z_{2.5\%}} \cdot \overbrace{\sqrt{\frac{0.25}{100}}}^{\sqrt{s^2/100}} \right]$,
simplifying to approximately $[0.44, 0.64]$.

Note: We can't be certain that the parameter p lies between 0.44 and 0.64. But if we polled 100 randomly chosen people repeatedly, say 1000 times (this amounts to observing 1000 distinct realizations of $X_1, \dots, X_{100}$), out of the 1000 confidence intervals that we get, about 95% of them would contain the true value of p .

Let's also calculate a lower one-sided 95% CI for p !

$$\leadsto z_{5\%} = 1.65 \text{ (from a table)}$$

$$\text{So: } \left[\bar{x} - z_{5\%} \cdot \frac{s}{\sqrt{n}}, \infty \right)$$

$$\Rightarrow [0.46, \infty) \text{ is a lower one-sided 95\% CI for } p.$$

Since we know $p \leq 1$ (as it is a proportion), this becomes $[0.46, 1]$.

Small Sample Sizes:

So far, we've built CI's for the mean of

- Any population distribution
- With known or unknown variance

|| • Only when the sample size is large ($n > 30$ for known variance, $n > 40$ o/w)

↑ We will be able to remove this restriction, but only by adding another - our population will need to be normal.

⊛ Prop. (Distribution of $\bar{X}_n$ for Normal Pop. w/ known variance):

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, with known σ^2 .

Then, for any n :

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0, 1), \text{ exactly.}$$

Proof: This follows from the fact that a linear combination of normal random variables will always be normal.

Since we are dividing by σ , a constant, this holds.

Note: 3 differences from the CLT:

- i) This holds for any n
- ii) The resulting distribution is exact
- iii) This applies only when X_i 's are iid Normal

So, we can now build CI's for the mean of a normally distributed population when the variance is known, for any sample size.